

Circuit Theory

Chapter 14 Frequency Response

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Frequency Response

Chapter 14

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- 14.2 Transfer Function
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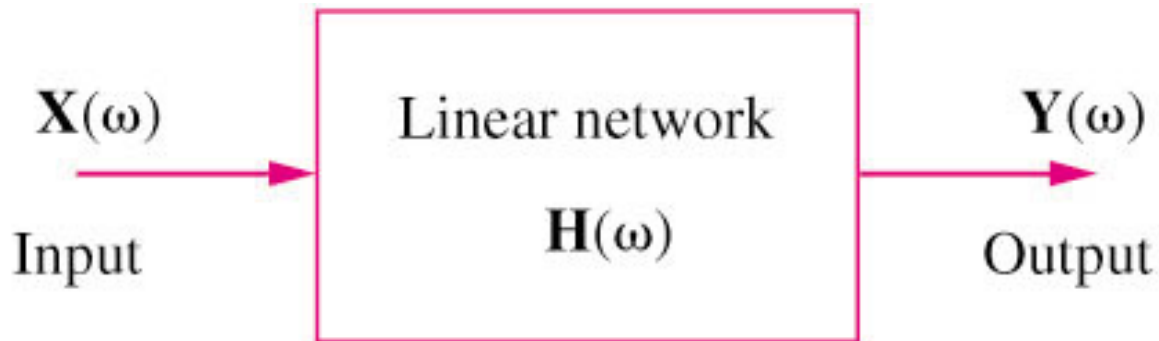
14.1 Introduction (1)

What is FrequencyResponse of a Circuit?

It is the variation in a circuit's behavior with change in signal frequency and may also be considered as the variation of the gain and phase with frequency.

14.2 Transfer Function (1)

- The transfer function $H(\omega)$ of a circuit is the **frequency-dependent ratio** of a phasor output $\underline{Y}(\omega)$ (an element voltage or current) to a phasor input $\underline{X}(\omega)$ (source voltage or current).



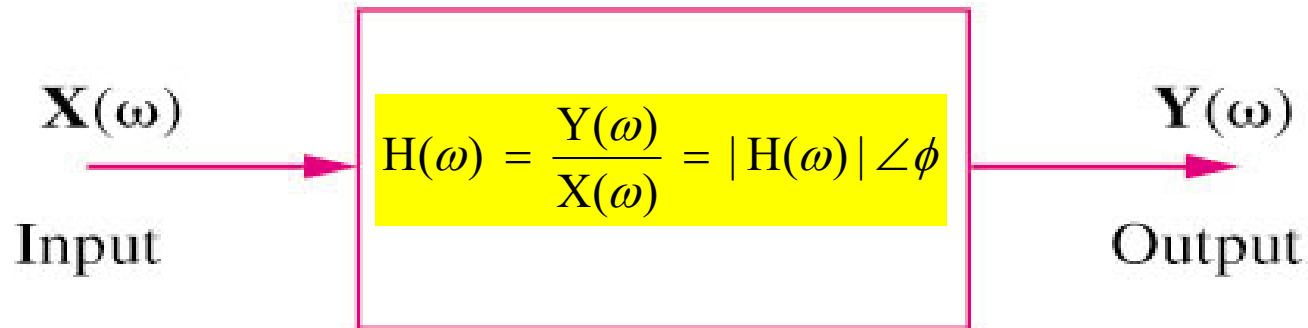
$$H(\omega) = \frac{\underline{Y}(\omega)}{\underline{X}(\omega)} = |H(\omega)| \angle \phi$$

14.2 Transfer Function (2)

- Four possible transfer functions:

$$H(\omega) = \text{Voltage gain} = \frac{V_o(\omega)}{V_i(\omega)}$$

$$H(\omega) = \text{Transfer Impedance} = \frac{V_o(\omega)}{I_i(\omega)}$$



$$H(\omega) = \text{Current gain} = \frac{I_o(\omega)}{I_i(\omega)}$$

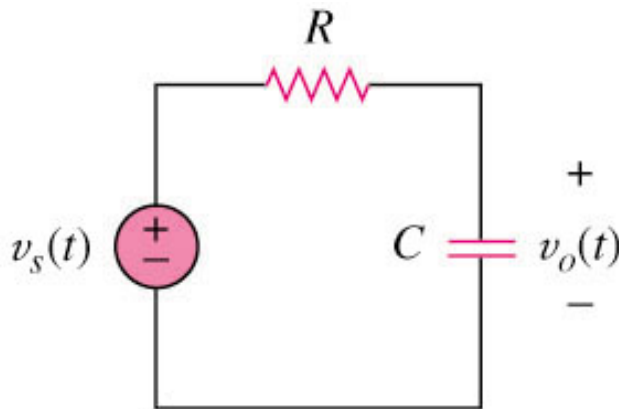
$$H(\omega) = \text{Transfer Admittance} = \frac{I_o(\omega)}{V_i(\omega)}$$

14.2 Transfer Function (3)

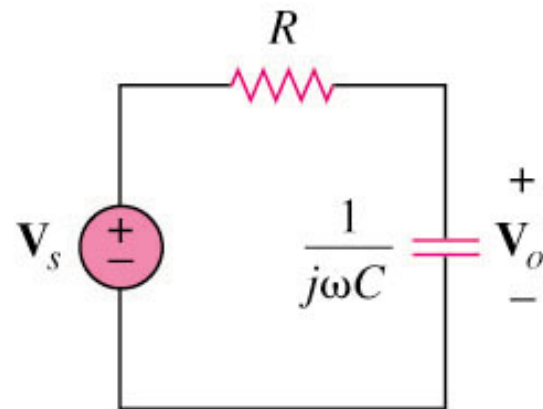
Example 1

For the RC circuit shown below, obtain the transfer function V_o/V_s and its frequency response.

Let $v_s = V_m \cos \omega t$.



(a)



(b)

14.7 Passive Filters (1)

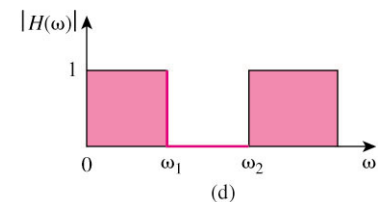
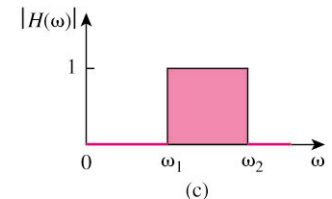
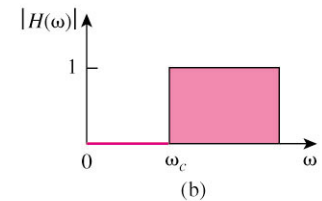
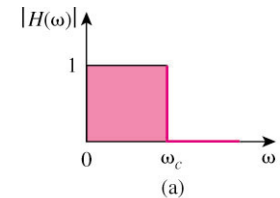
- A filter is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.
- **Passive filter** consists of only passive element R, L and C.
- There are four types of filters.

Low Pass

High Pass

Band Pass

Band Stop

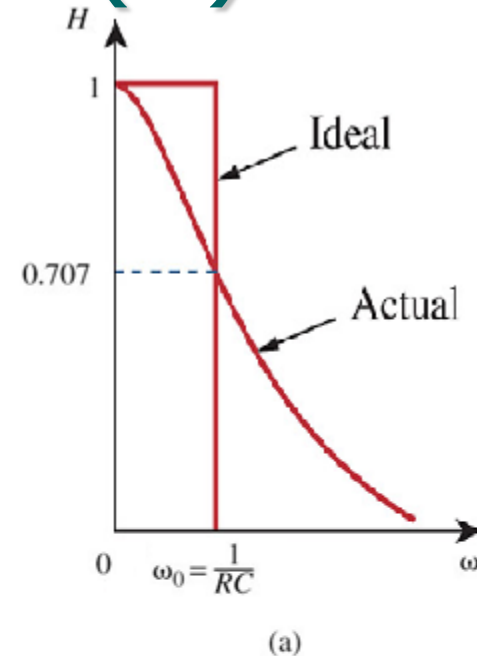
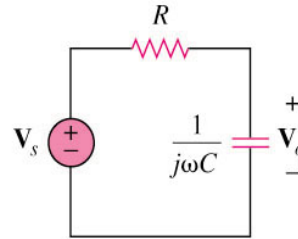


14.2 Transfer Function (4)

Solution:

The transfer function is

$$H(\omega) = \frac{V_o}{V_s} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{1 + j\omega RC}$$



The magnitude is

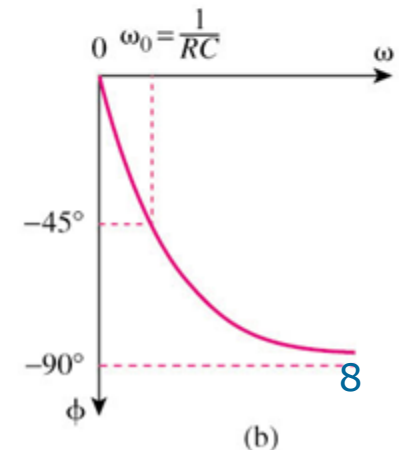
$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega / \omega_o)^2}}$$

The phase is

$$\phi = -\tan^{-1} \frac{\omega}{\omega_o}$$

$$\omega_o = 1/RC$$

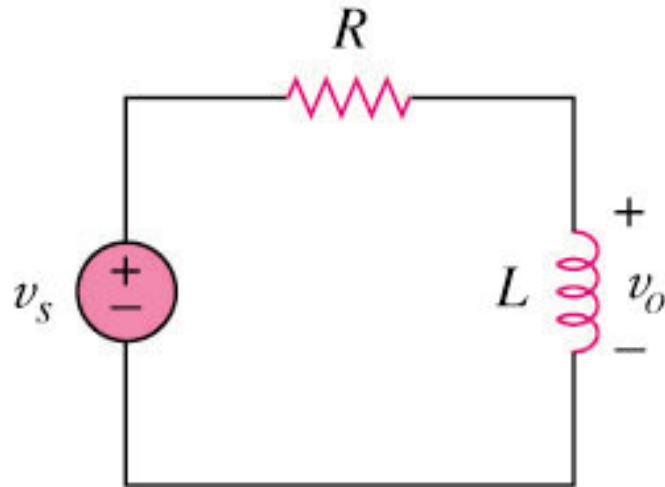
Low Pass Filter



14.2 Transfer Function (5)

Example 2

Obtain the transfer function V_o/V_s of the RL circuit shown below, assuming $v_s = V_m \cos \omega t$. Sketch its frequency response.

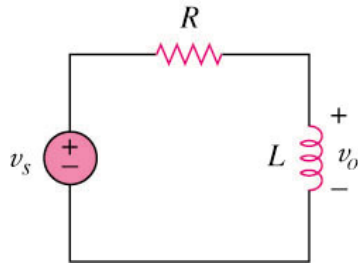


14.2 Transfer Function (6)

Solution:

The transfer function is

$$H(\omega) = \frac{V_o}{V_s} = \frac{j\omega L}{R + j\omega L} = \frac{1}{1 + \frac{R}{j\omega L}}$$



The magnitude is

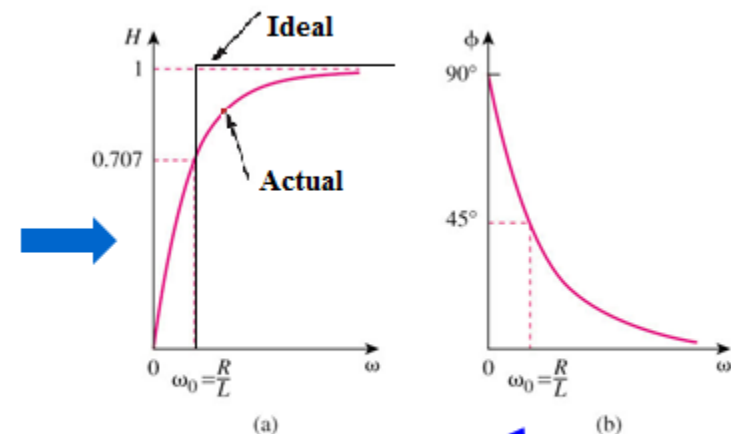
$$H(\omega) = \frac{1}{\sqrt{1 + \left(\frac{\omega_o}{\omega}\right)^2}}$$

The phase is

$$\phi = \angle 90^\circ - \tan^{-1} \frac{\omega}{\omega_o}$$

$$\omega_o = R/L$$

High Pass Filter



14.7 Band-Pass Filter (2)

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$$H(\omega) = \frac{R}{R + j\omega L + 1/j\omega C}$$

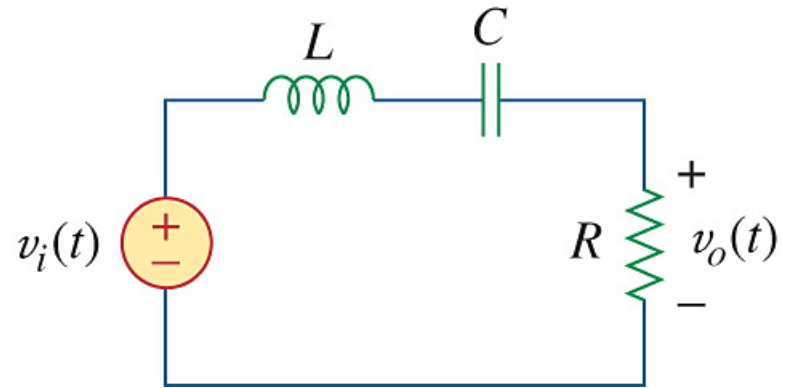
$$|H(\omega)| = \frac{\omega RC}{\sqrt{(1 - \omega^2 LC)^2 + (\omega RC)^2}}$$

$$\omega_o = 1/\sqrt{LC}$$

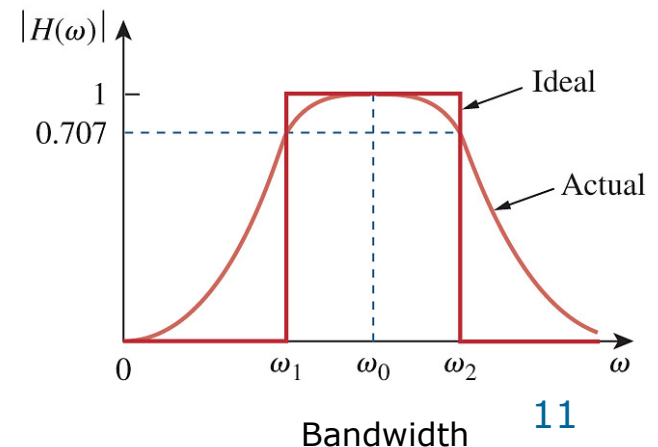
$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\text{Bandwidth} = \omega_2 - \omega_1 = R/L$$

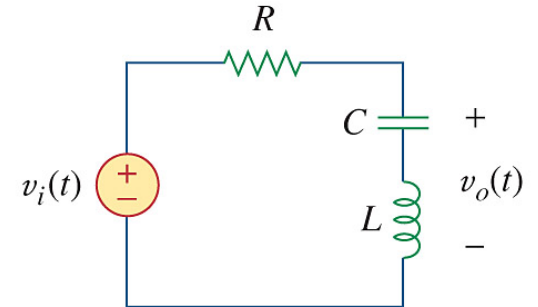


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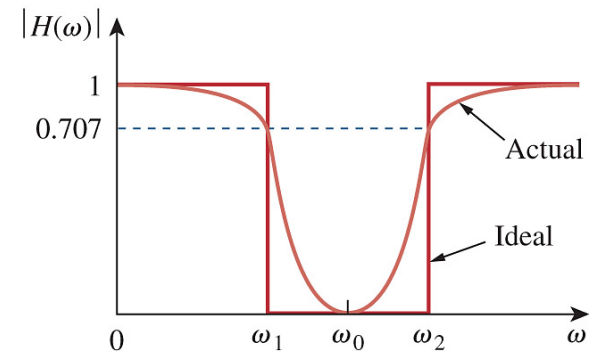


14.7 Band-Stop Filter (2)

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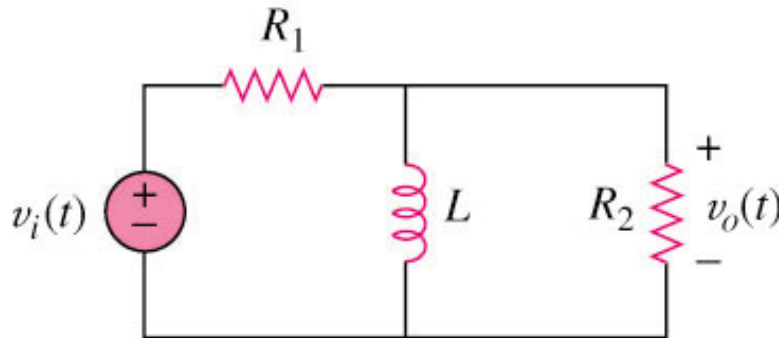
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14.7 Passive Filters (2)

Example 5

For the circuit in the figure below, obtain the transfer function $V_o(\omega)/V_i(\omega)$. Identify the type of filter the circuit represents and determine the corner frequency. Take $R_1=100\Omega = R_2$ and $L = 2\text{mH}$.



Answer: $\omega = 25 \text{ krad/s}$